

## Day 3 — How Many $3 \times 3$ Magic Squares?

**Module:** Magic Squares — Bhadrageṇita · **Band:** Senior · **Time:** 45 min

### Learning objective

By the end of this lesson, students will: count the equivalence classes (up to symmetry).

### Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

### Vocabulary introduced today

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa’s algorithm for odd-order squares”)

### Lesson flow

#### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

#### Core (20 min)

1. **How many  $3 \times 3$  magic squares are there?** Counting carefully:
  - 1 fundamentally different square (up to symmetry).
  - 8 squares including rotations ( $\times 4$ ) and reflections ( $\times 2$ ).
2. **Demonstrate the symmetries.** Start with 2-7-6 / 9-5-1 / 4-3-8. Rotate  $90^\circ \rightarrow$  4-9-2 / 3-5-7 / 8-1-6. Rotate again  $\rightarrow$  8-3-4 / 1-5-9 / 6-7-2. Etc. Also reflect.
3. **Define “essentially different.”** Two squares are equivalent if one is obtained from the other by rotation or reflection. The count of essentially distinct  $n \times n$  magic squares is one of the harder combinatorial problems.
4. **Higher orders.** Number of essentially-different magic squares: 1 (for  $n=3$ ), 880 ( $n=4$ ),  $\sim 275$  million ( $n=5$ ). Source: combinatorial computer enumeration (Frenicle, Schinzel).
5. **Class exercise.** Each pair writes out all 8 symmetric variants of the  $3 \times 3$  from Day 1. Verify each is still a magic square.

#### Activity (15 min)

A scaled version of *Class  $3 \times 3$  Magic-Square Tournament* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces How Many  $3 \times 3$  Magic Squares?.

### Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

### Homework

(See `homework.md` for today’s task.)

### Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

### Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).